

Node.js Lab: Express.js Exception Boundaries

Duration : 30 Min.

Scenario

An Express.js web service crashes unexpectedly in production because async route handlers lack proper error catch boundaries, and global unhandled promise rejections are not intercepted.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Set up global process exception boundaries for `uncaughtException` and `unhandledRejection` events in `server.js` and wrap Express async endpoints in error handlers.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Global <code>uncaughtException</code> and <code>unhandledRejection</code> event listeners are registered	5 Marks
TC2	Application error boundaries route async exceptions safely	5 Marks

Total Score: 10 Marks